§17. Pure resolution.

Setting: U f.g. graded R-mod

$\mathbb{F}$: graded free res

$c_{i,p}$: number of $R(-p)$ in $\mathbb{F}_i$

$t_i := \min\{p \mid c_{i,p} \neq 0\}$ $\quad T_i := \max\{p \mid c_{i,p} \neq 0\}$

Def. $\mathbb{F}$ is pure

Proposition 17.9. *Let* **F** *be a graded free resolution of U. The resolution is q-linear if and only if $F_0 = R(-q)^{c_{0,q}}$ and the entries in the differential matrices are linear forms. In particular for $q = 0$, we have that* **F** *is linear if and only if $F_0 = R^{c_{0,0}}$ and the entries in the differential matrices are linear forms.*

pf. $F_i = R(-p-i)^{c_{i,p+i}} \longrightarrow R(-p-i+1)^{c_{i,p+i-1}}$

Def. Fix p, Let $J \subseteq m^2$ be a graded ideal in S.

S/J has the property N_p of

$b^S_{i,i+1+j}(S/J) = 0$ for all $i \leq p$, $j > 0$.

$N_0 \iff$ projective normal

$N_1 \iff N_0 + J$ generated by quadric

$F_1 = R(-2)^{c_{1,2}}$

§18. Regularity.

Def. U R-mod

$\mathrm{reg}_R(U) = \max\{j \mid b^R_{i,i+j}(U) \neq 0 \text{ for some } i\}$.

$F_i = \bigoplus_p R(-p)^{c_{i,p}}$ $\qquad b_{i,i+j} = c_{i,i+j}$

prop. (1) U has a q-linear res, then $\mathrm{reg}(U) = q$.

(2) $\mathrm{reg}(I) = \mathrm{reg}(S/I) + 1$

$$\begin{tikzcd} F_i \arrow[r] & F_{i-1} \arrow[r] & \cdots & F_1 \arrow[r] \arrow[dr] & F_0 \arrow[r] & S/I. \\ & & & & I \arrow[u] & \end{tikzcd}$$

($F_0 = S$)

Thm. J graded artinial ideal in S

Set $q = \max\{i \mid (S/J)_i \neq 0\}$

Then $\mathrm{reg}(S/J) = q$.

pf. J artinial $\Rightarrow q < \infty$.

① $\mathrm{Tor}^S_{i,i+j}(S/J, k) = H_{i,i+j}(S/J \otimes \mathbb{K})$ ← koszul resolution of k

$\mathrm{Tor}^S_{i,i+j} = b_{i,i+j}$

Since $(S/J \otimes \mathbb{K})_{i+j} = 0$ for $j > q$.

$\Rightarrow b_{i,i+j} = 0 \quad j > q$.

$\Rightarrow \mathrm{reg}(S/J) \leq q$.

② $b_{n,n+q} = \mathrm{Tor}^S_{n,n+q}(S/J, k)$

$\mathrm{Tor}^S_{n,n+q}(S/J,k) = H_{n,n+q}(S/J \otimes \mathbb{K}) = \ker(d_n)$

$\overset{?}{=} \{f \in (S/J)_{n+q} \mid mf = 0\} \neq 0$?

$b_{n,n+q} \neq 0 \Rightarrow \operatorname{reg}(S/J) \geq q$. $\quad \mathfrak{m} = (x_1, \cdots, x_n)$

Pf of 14.11

that $\mathrm{H}_n(S/I \otimes \mathbf{K})$ is the kernel of the homomorphism

$$S/I \otimes K_n \cong S/I \to S/I \otimes K_{n-1}$$

$$f\, e_1 \wedge \cdots \wedge e_n \mapsto \sum\nolimits_{1 \leq p \leq n} (-1)^{p+1} x_p f\, e_1 \wedge \cdots \wedge \widehat{e}_p \wedge \cdots \wedge e_n\,,$$

where $f \in S/I$. It follows that $\mathrm{H}_n(S/I \otimes \mathbf{K}) = \{ f \in S/I \,|\, \mathbf{m} f = 0 \}$, which is the socle of S/I. $\square$

Q.

Proposition 18.6. *Suppose that*

$$0 \to U \to U' \to U'' \to 0$$

is a short exact sequence of graded finitely generated R-modules with homomorphisms of degree 0. Then

(1) *If* $\operatorname{reg}(U') > \operatorname{reg}(U'')$, *then* $\operatorname{reg}(U) = \operatorname{reg}(U')$.
(2) *If* $\operatorname{reg}(U') < \operatorname{reg}(U'')$, *then* $\operatorname{reg}(U) = \operatorname{reg}(U'') + 1$.
(3) *If* $\operatorname{reg}(U') = \operatorname{reg}(U'')$, *then* $\operatorname{reg}(U) \leq \operatorname{reg}(U'') + 1$.

pf. (1) $\to \operatorname{Tor}^R_{i+1}(U'', k)_j \to \operatorname{Tor}^R_i(U, k)_j$

$\to \operatorname{Tor}^R_i(U', k)_j \to \operatorname{Tor}^R_i(U'', k)_j \to \cdots$.

$\operatorname{reg}(U') > \operatorname{reg}(U'')$

① Claim. $\operatorname{Tor}^R_i(U', k)_{i+\operatorname{reg}(U')} \neq 0$

claim $\Rightarrow \operatorname{reg}(U) \geq \operatorname{reg}(U')$

pf of claim.

$j = i + \operatorname{reg}(U')$

$\to \operatorname{Tor}^R_i(U, k)_j \to \operatorname{Tor}^R_i(U', k)_j \neq 0$

$\neq 0 \qquad \to \operatorname{Tor}^R_i(U'', k)_j \neq 0$

②. Suppose $\operatorname{reg}(U) > \operatorname{reg}(U')$

$\operatorname{Tor}^R_i(U, k)_{i+\operatorname{reg}(U)} \neq 0$

$j = i + \mathrm{reg}(U)$

$$\underset{\neq 0}{\mathrm{Tor}^R_{i+1}(U'', k)_j} \to \underset{\neq 0}{\mathrm{Tor}^R_i(U, k)_j} \to \underset{= 0}{\mathrm{Tor}^R_i(U', k)_j}.$$

$\mathrm{reg}(U) > \mathrm{reg}(U') > \mathrm{reg}(U'')$

$\Rightarrow \mathrm{reg}(U'') \leq \mathrm{reg}(U) - 2 \Rightarrow \mathrm{Tor}^R_{i+1}(U'', k)_j = 0.$

Contradict

$\Rightarrow \mathrm{reg}(U) = \mathrm{reg}(U')$

$\square$